

UNIT IV
ASSIGNMENT

Class #3: Gram Schmidt process:

1) Let $w = \text{span} \{x_1, x_2\}$, where $x_1 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$ and $x_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. Construct an orthogonal basis $\{v_1, v_2\}$ for w .

$$\text{ANS: } v_1 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

2) Let $x_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $x_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $x_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, then $\{x_1, x_2, x_3\}$ is clearly linearly

independent and thus is a basis for the subspace w of R^4 . Construct an orthogonal basis for w .

$$\text{ANS: } v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} -\frac{3}{4} \\ \frac{1}{4} \\ \frac{1}{4} \\ \frac{1}{4} \end{pmatrix}, v_3 = \begin{pmatrix} 0 \\ \frac{-2}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix}$$

3) Apply the gram Schmidt's orthogonalization for $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$Q = \begin{pmatrix} \frac{1}{2} & \frac{3}{\sqrt{12}} & 0 \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & \frac{-2}{\sqrt{6}} \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{6}} \\ \frac{1}{2} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{6}} \end{pmatrix}$$